

Seonghyun Jin

Integrated M.S./Ph.D. Student | KAIST Graduate School of AI

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RESEARCH INTERESTS

Reconstructing and generating the real 3D world; streaming 3D reconstruction; geometric perception; Geometric Positional encoding; camera-controlled video generation.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Integrated M.S./Ph.D. in Artificial Intelligence

– Advised by Prof. Jong Chul Ye at BISPL.

Mar. 2025 – Present

Daejeon, Republic of Korea

Seoul National University (SNU)

B.S. in Computer Science; B.A. in Economics, College of Liberal Studies

– GPA: 3.68/4.3, *Cum Laude*.

Mar. 2020 – Feb. 2025

Seoul, Republic of Korea

PUBLICATIONS

P: preprint, C: conference, W: workshop. *Equal contribution. Citations: [Google Scholar](#), Sep. 11, 2026.

[P2] Adjoint Guidance Flow: Amortized Critic Guidance for VLA Policies

Jeongsol Kim, Youngjun Jun, Kyumin Choi, Youngmin Kim, **Seonghyun Jin**, Sunwoo Park, Jangho Park, Kwanyoung Kim, Jong Chul Ye

[arXiv preprint, 2026](#) | [arXiv](#) | [PDF](#)

[P1] Dynamic Manipulation with World-Action Models via Counterfactual Planning

Sunwoo Park*, Wonbin Lee*, **Seonghyun Jin***, Youngmin Kim*, Jangho Park, Jong Chul Ye

[arXiv preprint, 2026](#) | [arXiv](#) | [PDF](#) | [Project](#)

[C3] CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation

Seonghyun Jin*, Youngmin Kim*, Sunwoo Park*, Jong Chul Ye

[NeurIPS 2026](#). Also accepted to [3D in the Era of World Models](#), [ECCV 2026 Workshop](#) | [arXiv](#) | [PDF](#) | [Workshop](#)

[C2] FILT3R: Latent State Adaptive Kalman Filter for Streaming 3D Reconstruction

Seonghyun Jin, Jong Chul Ye

[ECCV 2026](#) | [arXiv](#) | [PDF](#) | [Code](#) | [Project](#) | 2 citations

[C1] AI-Based Mental Health Assessment for Adolescents Using Their Daily Digital Activities

Do Hyung Kim, Joonsung Lee, Taehwi Lee, Soeun Baek, **Seonghyun Jin**, HaEun Yoo, Youngeun Cho, Seonghyeon Park, Kwangsu Cho, Chang-Gun Lee

[IEEE DSAA, 2024](#) | [PDF](#) | 10 citations

ACADEMIC EXPERIENCE

RUBIS Lab, Seoul National University

Undergraduate Research Intern

– Contributed to adolescent mental health digital phenotyping research through feature engineering and data collection support; co-authored an IEEE DSAA 2024 publication.

Sep. 2022 – Jun. 2024

Seoul, Republic of Korea

LANGUAGES

Korean (native); English (fluent, TEPS 538, TOEIC 980).

TECHNICAL SKILLS

Programming Languages Python, C, C++

Deep Learning Frameworks PyTorch

Computer Vision Multi-view geometry, 3D reconstruction, camera pose and depth estimation

Training Experience Deep learning training, ablation studies, experiment management, evaluation pipelines

REFERENCE

Professor Jong Chul Ye

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